

$$x_i = ih, \ y_j = jh, \ h = 1/N,$$

$$\frac{4u_{ij} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}}{h^2} = f_{ij},$$

$$A\mathbf{u} = \mathbf{f}, \qquad A = A^T > 0,$$

$$\mathbf{r} = \mathbf{f} - A\mathbf{u}_h,$$

$$\|u_h-u\|_\infty=O(h^2) \quad \text{for this smooth solution.}$$